

**REGULARIZING CONTROL INVARIANCE:
CONTROL BARRIER FUNCTIONS AND WHERE TO FIND THEM**

by

Victor Freire

A.S., Madison Area Technical College, 2017

B.S., University of Wisconsin–Madison, 2020

M.S., University of Wisconsin–Madison, 2022

A dissertation submitted to the
Faculty of the Graduate School of the

UNIVERSITY OF COLORADO BOULDER

in partial fulfillment of the requirements for the degree of

Doctor of Philosophy

Department of Electrical, Computer & Energy Engineering

2026

Committee:

Marco M. Nicotra, Chair

Emily Jensen

Sriram Sankaranarayanan

Sam Coogan

Hamid R. Ossareh

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ACKNOWLEDGMENTS

I thank my advisor, Marco Nicotra, for the insightful discussions and collaborative work that shaped the material in this dissertation. Most of all, however, I thank Marco for showing me that compassion, kindness, and humor are essential qualities of a great mentor.

I thank my dissertation committee: Emily, Sriram, Sam, and Hamid, for finding time in their busy schedules to provide diverse and invaluable insight during the final stages of my doctoral studies.

I thank Lori for being the best graduate program advisor in the universe.

I thank the National Science Foundation for providing financial security during my time as a Ph.D. student. This dissertation is based upon work supported by the CMMI-DCSD Award #2411667. Any opinions, findings, and conclusions or recommendations expressed are my own and do not necessarily reflect the views of the National Science Foundation.

I thank my labmates Addie, Izzy, Lily, Simon, Sam, Gabe, and Yaashia, who were most present when I wrote this, as well as all those who came and went over the years, for making my time at CU Boulder both engaging and enjoyable.

I thank my ultimate Frisbee community for making Mondays a day to look forward to instead of loathsome routine.

I thank my American family for always keeping a seat open at their table for me.

Agradezco a mis amigos de España por brindarme una ventana a mi infancia y evitar que mi español se oxide.

Agradezco a mi padre por su confianza constante en mí, su firme apoyo y por sentirse tan orgulloso de cada paso de mi camino.

Agradezco a mi hermano, Mario, por su camaradería de toda la vida, las risas y su apoyo incondicional, sin importar la distancia que nos separe.

I thank my partner, Tierney, for her unconditional love, for planning every one of our trips, and for all the walks and fetch with Titi. I can't wait for the many adventures and memories still ahead of us.

FOREWORD

At the time of writing this dissertation, the rapid development of artificial intelligence (AI) is changing our relationship with autonomous technology. We are entering an era where the volume of AI-driven decisions often outpaces our ability to verify their safety. While modern AI can make highly optimized decisions, it often operates as a closed box lacking formal guarantees. When human lives or critical infrastructure depend on these systems, empirical trust is insufficient.

The constrained control field offers a blueprint for how we might begin to address this. Through mathematical properties like control invariance and constructs like control barrier functions (CBFs), we can establish certificates of safety: proof that a dynamical system will never violate its imposed boundaries, regardless of the underlying complexity.

The research in this dissertation explores the connections between control invariance and CBFs. Moreover, it leverages tools from the reference governor literature to find CBFs. While artificial intelligence is not explicitly discussed in the chapters that follow, the theoretical foundations developed herein provide the rigorous mathematical groundwork necessary to one day guarantee the safety of AI-driven autonomous systems.



*Original artwork by Michele Carminati,
commissioned by Marco M. Nicotra.*

ABSTRACT

The control invariance property of sets was characterized in terms of the Bouligand contingent cone by Nagumo in 1942. Under standard regularity and transversality conditions, the expression for the contingent cone simplifies significantly at the boundary. This dissertation demonstrates how control barrier functions extend this characterization to the entire set beyond its boundary: a process termed regularization. In doing so, the dissertation exposes the geometric conditions under which control barrier functions exist. These existence conditions serve as the safety equivalent to the converse Lyapunov theorems for stability. Finally, to bridge theory and application, the dissertation highlights how tools from the reference governor literature can be used to obtain control barrier functions.

The complete dissertation will be available once all associated research is published.